



Basic Principles of Medicine 1

Module: Musculoskeletal System | IMCQ4

IMCQ 4

*School of Medicine,
St. George's University*

1. A 30-year-old woman comes to the physician for pelvic assessment at 38 weeks of gestation to evaluate the likelihood of successful vaginal delivery. The following measurements were within normal limits

- ¹. the distance between the sacral promontory and the superior margin of the pubic symphysis,
- ². the distance between the coccyx and
- ³. the inferior margin of the pubic symphysis and the distance between the ischial spines is within normal limits.

The distance between the sacral promontory and the inferior margin of the pubic symphysis is reduced. Which of the following best describes the measurement that is reduced in this patient?

- A. Conjugate AP
- B. Conjugate diagonal
- C. Sagittal Outlet
- D. True Conjugate
- E. Interspinous (bispinous diameter)

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SOM.MKII.BPM1.1.MSK.1.ANAT.1278 Describe the following dimensions of the pelvis: pelvic inlet, conjugate (antero-posterior) diameter, interspinous distance and pelvic outlet.

2. A 41-year-old woman comes to the physician because of persistent right buttock pain radiating to the posterior thigh. The pain worsens when she attempts to sit cross-legged or externally rotate her hip while standing. Physical examination shows weakness in the lateral rotation of the hip when extended and weakness in the abduction of the hip when flexed. Further testing suggests injury to a muscle that exits the pelvis through the greater sciatic foramen and is innervated by branches of the sacral plexus. Which of the following nerve roots provides innervation to the affected muscle?

- A. L4 – S1**
- B. L5 – S2**
- C. S1 – S2**
- D. S1 – S3**
- E. S2 - S4**

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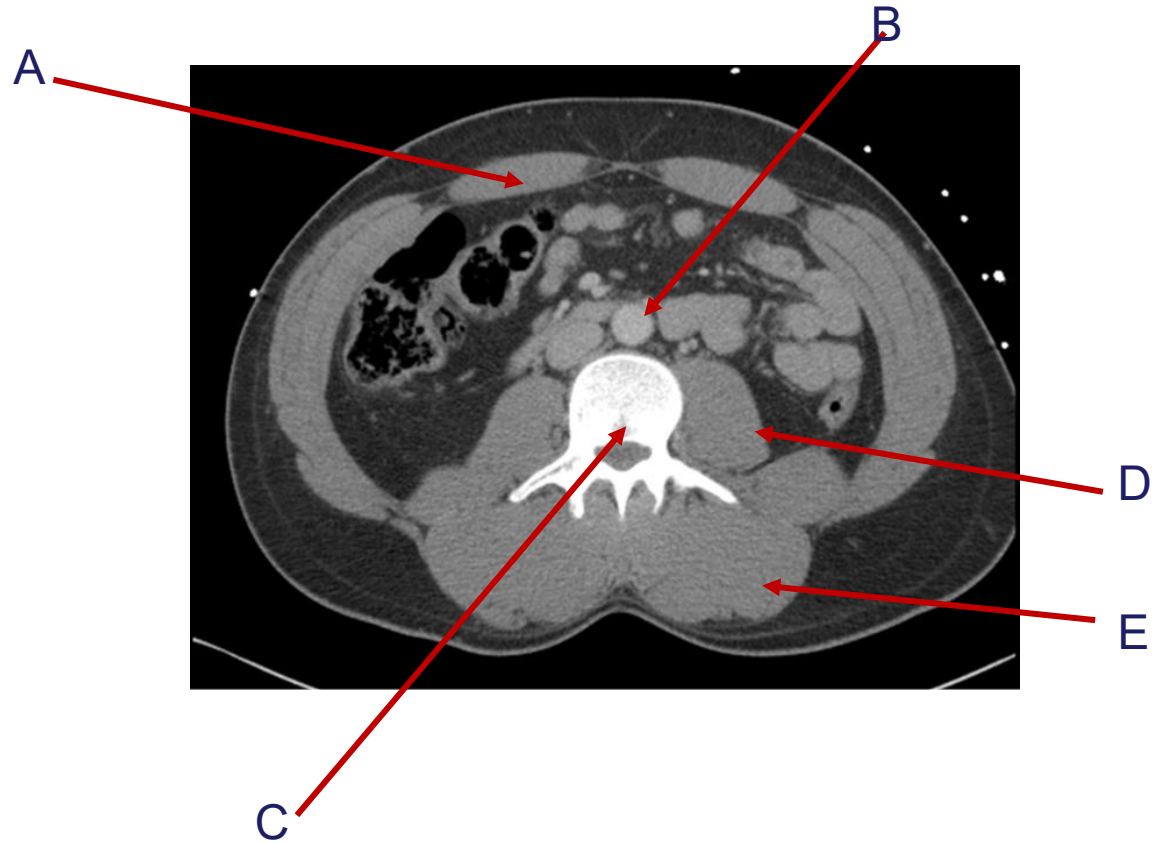
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SOM.MK.II.BPM1.1.MSK.1.ANAT.1288

"List the terminal branches of the sacral plexus and their distribution"

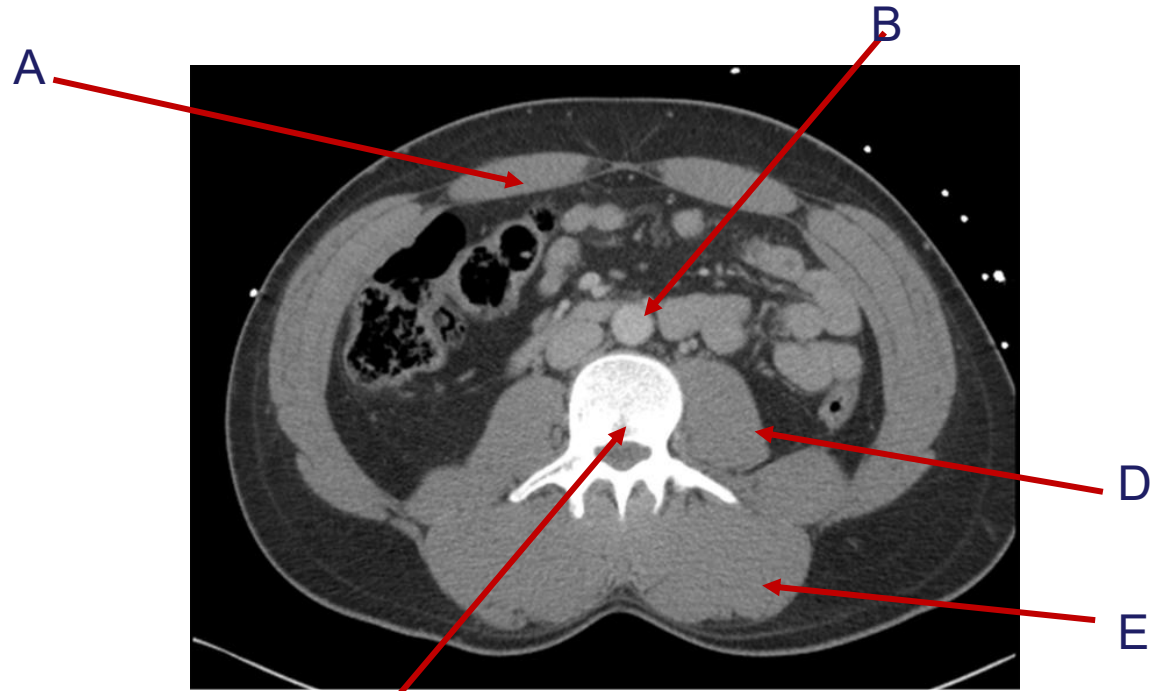
3. A 57-year-old man comes to the emergency department because of a sudden onset of severe, tearing abdominal pain radiating to his back. He has a 30-year history of hypertension and a 40-pack-year smoking history. His blood pressure is 92/54 mmHg, and pulse is 118/min. An urgent contrast-enhanced CT scan of the abdomen is performed, and the patient is diagnosed with a ruptured abdominal aortic aneurysm. Which of the following labels correctly identifies the structure that is most likely affected?

- A. A
- B. B
- C. C
- D. D
- E. E



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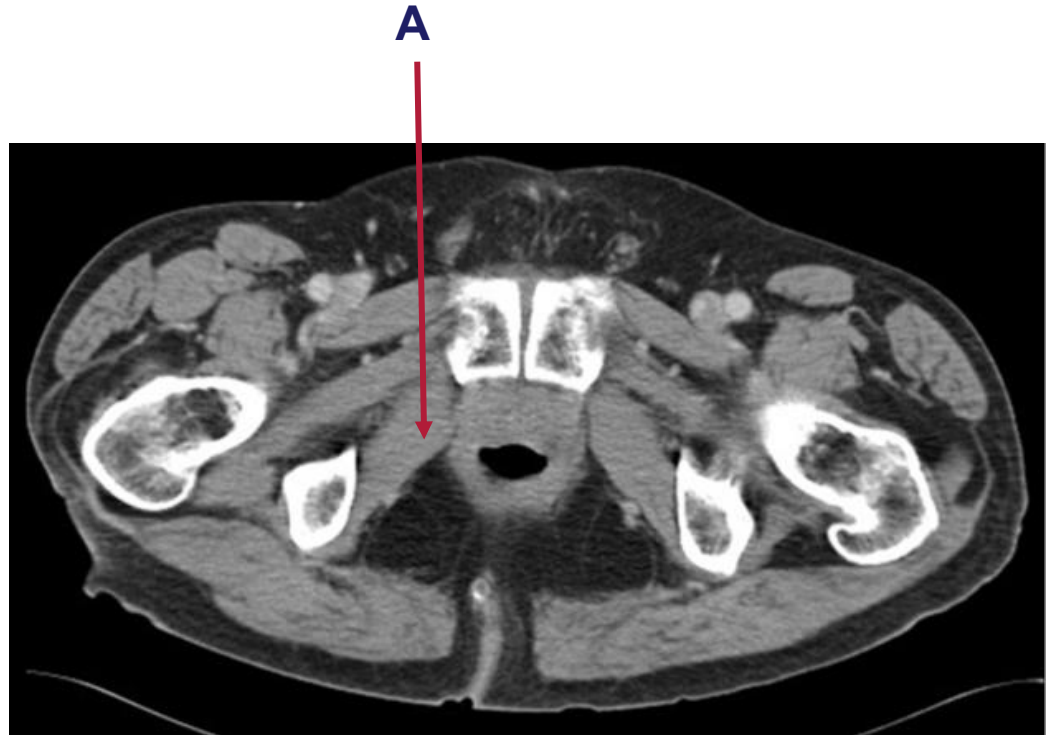
C
List the parietal and visceral branches of the abdominal aorta and use this information to identify them on cadavaric specimen and imaging studies.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1268

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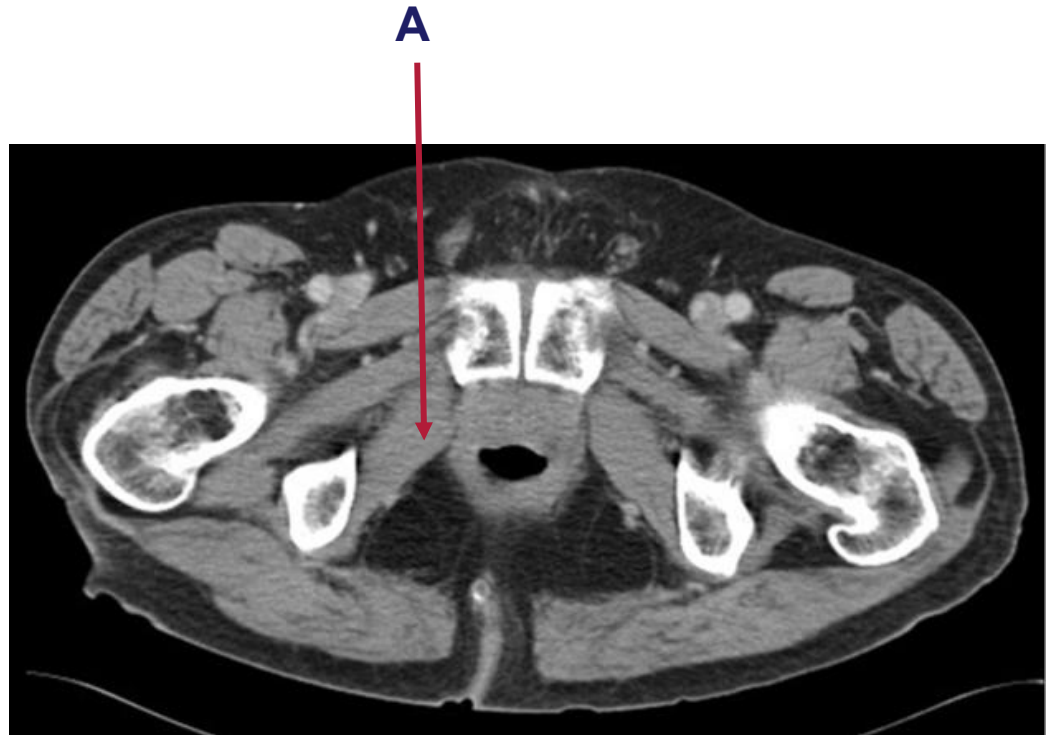
4. A 46-year-old man comes to the physician because of a 6-month history of chronic pelvic pain that worsens with prolonged sitting. He denies any trauma, fever, or urinary symptoms. Vital signs are within normal limits. Physical examination shows tenderness on deep palpation of the posterior pelvic region. An axial CT scan of the pelvis is obtained. Injury to the muscle labelled A will most likely impair which of the following movements of the hip?

- A. Adduction
- B. Medial rotation
- C. Lateral rotation
- D. Flexion
- E. Extension



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5. During the postnatal assessment of a boy, an abnormality is observed in his right hand as shown in the image. Which of the following is the most likely embryological explanation responsible for this abnormality?



- A. Amniotic band construction
- B. Disruption of the anterior-posterior pattern
- C. Failure of development of the notches between the digital rays
- D. Failure of webs to degenerate between 2 or more digits
- E. Failure of development of one or more digital rays
- F. Suppression of limb bud development in the 4th week

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SOM.MK.II.BPM1.1.MSK.1.ANAT
.1345 Explain the formation of the following: congenital talipes (clubfoot), amelia / meromelia, cutaneous syndactyly, osseous syndactyly, brachydactyly, polydactyly, ectrodactyly (cleft hand or partial adactyly) and limb deformities caused by amniotic bands.

6. A 35-year-old woman is brought to the emergency department because of a 2-hour history of a blow to the lateral aspect of her knee joint. Physical examination shows difficulty extending his digits and big toe. An x-ray of her knee joint is shown. Which of the following additional symptoms will most likely be present?



<https://epos.myesr.org/posterimage/essr/essr2015/130990/medialgallery/635494?deliveroriginal=1>

- A. Difficulty plantarflexing the ankle joint and numbness in the 1st web space
- B. Loss of sensation on the medial arch of the foot and difficulty everting the ankle joint
- C. Foot drop and numbness on the entire dorsum of the foot
- D. Loss of sensation on the lateral arch of the foot and difficulty inverting the ankle joint
- E. Difficulty plantarflexing the ankle joint and numbness in the 2nd web space.

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1322

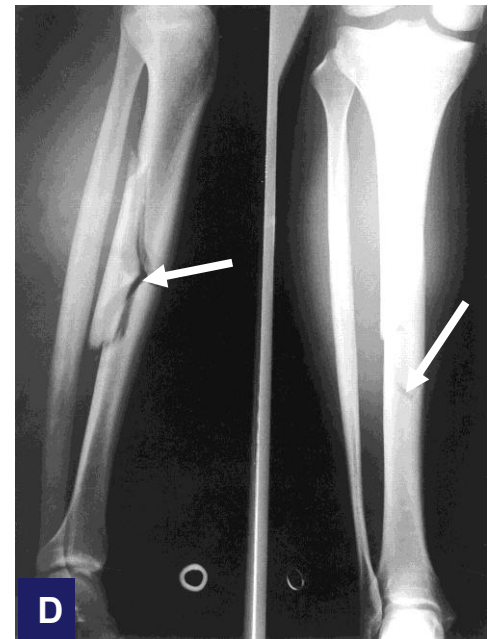
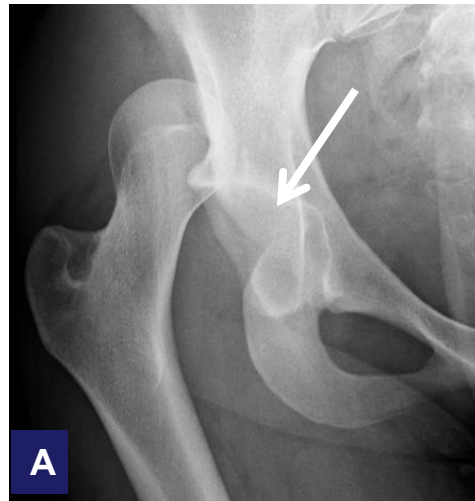
Explain the consequence of a lesion of the common fibular nerve at the neck of the fibula.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1323

Explain the consequence of a lesion of the superficial and deep fibular nerves just distal to their origin.

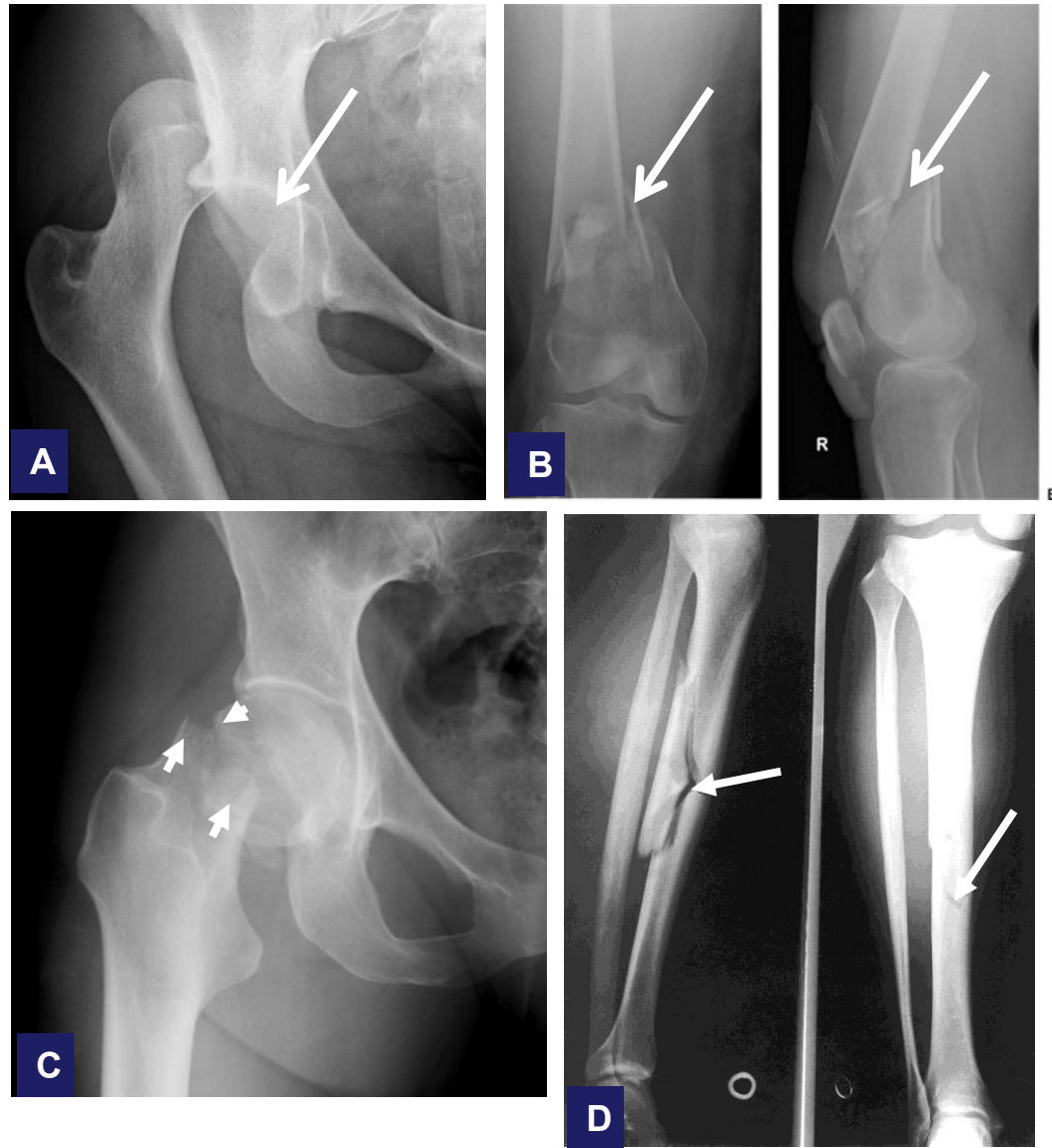
7. A 35-year-old man is brought to the emergency department by paramedics after being involved in a motor vehicle collision. Physical examination shows decreased popliteal, posterior tibial and dorsalis pedis pulses in his right lower limb. All other pulses are normal. Which of the following is most likely expected in a radiograph of his right lower limb?

- A. A
- B. B
- C. C
- D. D



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- B. **B**
- C. C
- D. D



SOM.MK.II.BPM1.1.MSK.1.ANAT.1297 Describe the clinical significance of an intracapsular versus an extracapsular fracture of the femur and identify on imaging studies.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1496 Identify femoral fractures in imaging studies and discuss likely clinical signs as symptoms as a result of these fractures

8. A 24-year-old woman comes to the physician because of a 1-month history of pain in the right knee. She is a basketball player and has been training extensively in preparation for a series of upcoming games. Physical examination shows no redness or swelling of the knee but there is point tenderness on palpation of the structure outlined on the bone image. Passive range of motion is normal but active movement against resistance produces pain at the same point. Which of the following movements would most likely produce this pain and which nerve(s) innervates the muscles producing this movement?



- A. Flexion of the knee joint and femoral nerve
- B. Extension of the knee joint and femoral nerve
- C. Extension of the hip joint and sciatic nerve
- D. Flexion of the hip joint and sciatic nerve
- E. Extension of the knee joint and sciatic nerve

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- E. Extension of the knee joint and sciatic nerve

SOM.MK.II.BPM1.1.MSK.1.ANAT.1304 1. List the muscle compartments of the thigh. 2. Describe the general attachments of the muscles of thigh region, their actions and neurovascular supply.

9. A 3-month-old boy is brought to the clinic for evaluation of congenital limb malformations. He was born at term after an uncomplicated pregnancy. Physical examination shows bilateral shortening of the upper limbs, missing digits, and hypoplastic scapulae. His skull, vertebral column, and rib development are normal. A mutation affecting mesodermal development is suspected. Which of the following embryologic structures is most likely disrupted in this patient?

- A. Paraxial mesoderm
- B. Lateral plate mesoderm
- C. Intermediate mesoderm
- D. Neural crest cells
- E. Ectoderm
- F. Endoderm

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1343

Describe the musculoskeletal development of the limbs from the lateral intraembryonic mesoderm and myotomes of the somites.



10. A 30-year-old woman comes to the physician because of a 1-week history of redness and swelling over the medial side of her right foot after scratching it on a rock during a trail run. Physical examination shows a 2-cm erythematous laceration in the skin over the medial arch of her right foot, just anterior to the medial malleolus. There is decreased sensation over the affected area. Which of the following best describes the cutaneous innervation and lymphatic drainage of the affected area?

- A. Sural nerve and popliteal lymph nodes
- B. Lateral plantar nerve and popliteal lymph nodes
- C. Saphenous nerve and vertical group of superficial inguinal lymph nodes
- D. Deep fibular nerve and horizontal group of superficial inguinal lymph nodes
- E. Medial plantar nerve and popliteal lymph nodes
- F. Superficial fibular nerve and vertical group of superficial inguinal lymph nodes

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1334

Predict the regional lymph nodes which will be most likely enlarged because of infection(s) in the lower limb.

11. A 65-year-old man comes to the physician because of a 2-month history of difficulty walking. Physical examination shows when standing with his right foot off the floor, his unsupported hip drops, giving a positive Trendelenburg sign. Muscle strength is 5/5 in both lower limbs except for mild weakness in right hip abduction. Reflexes and sensation are intact. Which of the following are most likely the roots of the nerve that innervates the affected muscles?

- A. S2 - S4
- B. L5 – S2
- C. S1 – S3
- D. L 2 – L3
- E. L2 - L4
- F. L4 –S1
- G. L4 – S3
- H. L4 – S2
- I. L1- L2

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- H. L4 – S2
- I. L1- L2

SOM.MK.II.BPM1.1.MSK.1.ANAT.1298 Describe the general attachments of the muscles of gluteal region, their actions and neurovascular supply.

12. A 45-year-old woman is brought to the emergency department because of a 3-day history of lower back pain. The pain radiates down her left buttock, posterior thigh, and calf to the lateral foot. Physical examination shows motor and sensory deficits. An MRI of her lumbar spine is shown. Which of the following deficits will most likely be found on physical examination?

- A. Diminished dorsiflexion of the ankle joint
- B. Loss of sensation on the skin over the inguinal ligament
- C. Diminished Achilles tendon reflex
- D. Loss of sensation on the medial side of the big toe
- E. Diminished patella reflex



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SOM.MKII.BPM1.1.MSK.1.ANAT.1115 Describe the process of disc herniation and its consequences in terms of nerve impingements paying particular attention to the differences between the cervical, thoracic and lumbar regions.

13. A 62-year-old man comes to the clinic because of cramping pain in his right calf and foot when walking more than one block. The pain is relieved by rest. He has a history of hypertension, hyperlipidemia, and smoking. Doppler ultrasound shows a complete occlusion of the right popliteal artery at the area indicated in the diagram. Which of the following arteries most likely provide collateral blood flow to the distal limb in this patient?

- A. Posterior tibial and anterior tibial
- B. Fibular and anterior tibial
- C. Inferior medial genicular and inferior lateral genicular
- D. Lateral femoral circumflex and descending genicular
- E. Descending genicular and medial femoral circumflex



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14. A 25-year-old man comes to the physician because of a 1-week history of difficulty in moving some of the joints of his lower limb. Three weeks ago, he sustained an injury to his right lower limb which was treated and has since healed. Physical examination shows he is able to stand on his toes and has loss of sensation on the plantar surface of the third digit. Which of the following motor deficits will most likely be found on physical examination?

- A. Difficulty everting the ankle joint
- B. Loss of flexion of the digits
- C. Difficulty extending the digits
- D. Loss of adduction of the big toe
- E. Loss of abduction the big toe

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1321

Explain the consequence of a lesion of the tibial nerve in the popliteal fossa.

15. A 25-year-old man comes to the emergency department after twisting his right ankle during a soccer game. He reports that his foot rolled inward when he landed awkwardly while running. He was unable to continue playing and has had significant swelling and pain since the injury. Physical examination shows marked swelling and ecchymosis over the ankle joint. Which of the following ligaments is most likely injured in this patient?

- A. Anterior tibiotalar
- B. Calcaneofibular
- C. Posterior talofibular
- D. Calcaneonavicular (spring)
- E. Tibiocalcaneal

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1326 Describe the ankle joint
its and ligamentous support

SOM.MK.II.BPM1.1.MSK.1.ANAT.1327 List the two ligaments
most commonly damaged during an inversion sprain.